Proceedings of the ASME 2022
International Mechanical Engineering Congress and Exposition
IMECE2022
October 30 - November 3, 2022, Greater Columbus Convention Center, Columbus, Ohio (O.H.), USA

IMECE2022-95121

MITIGATION OF THRUST DETERIORATION AT A HIGH FLAPPING FREQUENCY OF A 2D AIRFOIL IN FORWARDING FLIGHT CONDITIONS USING ASYMMETRIC FLAPPING STROKES

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ABSTRACT

In this article, the detrimental time-averaged thrust performance of a two-dimensional elliptic symmetrical flapping airfoil at very high flapping frequencies is addressed, and an attempt is made to enhance the thrust performance by implementing two separate flapping frequencies during the downstroke and upstroke of a flapping cycle thus making it asymmetric. Three different pivot locations, three effective angle of attack amplitudes, and a specific range of St are considered for our investigation, keeping the Reynolds number fixed as 5000. In general, the downstroke flapping frequency is always selected with a value higher than the critical Strouhal number, but the upstroke flapping frequency is kept fixed as the critical Strouhal number. It is worth highlighting that the observations are periodic and swapped suitably when the flapping stroke frequency selection is reversed in the flapping cycle. Asymmetric flapping stroke configurations yield more time-averaged thrust and non-zero lift than symmetric flapping stroke at thrust degrading flapping frequency range. Interestingly, the leading edge side pivot point produces positive lift when the implementation of a faster downstroke and slower upstroke configuration. At the same time, it's the opposite for the center and trailing edge pivot points. To understand these enhanced aerodynamic performances at high flapping frequencies, we have further analyzed the transient thrust and lift force profiles and their associated flow structures for all the pivot locations.

Keywords: Pivot point locations; Asymmetric flapping airfoil; Non-zero lift, Water tunnel experiments; Ansys Fluent.

NOMENCLATURE

A	Width of the wake	[m]
c	Chord length	[m]
C_T	Thrust co-efficient	--
C_L	Lift co-efficient	--
f_X	Flapping frequency	Hz
h_0	Heaving amplitude	[m]
$h(t)_X$	Instantaneous heaving function	--
L	Span of airfoil	[m]
Re	Reynolds number	--
St	Strouhal number	--
t	Physical dimensional time	[sec]
T	Thrust force	[N]
T'	Physical flapping cycle time	[sec]
U_∞	Free stream velocity	[m/s]
S	Planform area of the airfoil	[m^2]
ω_X	Angular flapping frequency	[rad/s]
$\theta(t)_X$	Instantaneous pitching function	--
θ_0	Pitching amplitude	(^{0})
ϕ	Phase angle	(^{0})
ν	Kinematic viscosity	[m^2/sec]
ρ	Density	[kg/m^3]

1. INTRODUCTION

In symmetric flapping airfoil studies, thrust performance reduces drastically at a high flapping frequency (or *St*) in a forward flight condition. But many Unmanned Aerial Vehicles operate in this high *St* range. Also, the time-averaged lift force generated in a cycle is negligible for symmetric flapping. This issue can be resolved by introducing asymmetricity in pitching motion by implementing two separate flapping frequencies during downstroke and upstroke.

2. LITERATURE REVIEW AND OBJECTIVES

[1] computationally investigated the effect of asymmetric oscillation of a NACA 0012 airfoil at a range of amplitude of oscillation and reduced frequencies at $Re = 1.35\times10^5$. They reported that the asymmetric ratio affects the instantaneous force coefficients and associated flow structures. It was also observed that faster downstroke could delay Leading Edge

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Vortex (LEV) formation and development and enhance the thrust performance. In a similar type of study, [2] investigated the effect of asymmetric sinusoidal oscillating motion on the propulsion performance of a pitching foil numerically and attempted to gain insight into whether the low thrust generated by pure pitching could be improved by asymmetric motion. The results show that the higher asymmetry ratio can induce the stronger reverse Von Karman vortex in the wake, leading to increased thrust but sacrificing propulsion efficiency. [3] numerically investigated the effect of symmetric and asymmetric pitching foil undergoing sinusoidal oscillation by varying thickness based on Strouhal number. They showed that the time-averaged thrust increases with an increase in asymmetry. A higher asymmetric constant value prepones drag to thrust transition. Apart from all these numerical studies, [4] conducted a numerical and experimental study analyzing the effect of asymmetric flapping of a tandem-foil system at various phase angles at Re = 5000. Increased asymmetric ratio and 180-degree phase difference create maximum thrust.

Additionally, we have reviewed some relevant studies on the two-dimensional flapping airfoils addressing the effects of the pivot locations on their propulsive performances. [5] and [6] have performed particle image velocimetry (PIV) and dye visualisation experiments on an SD7003 flapping airfoil at different pivot locations and flapping frequencies following a ramp-hold type pitch and sinusoidal heave motion. They found the evolution of the leading-edge vortices (LEVs) and wake structure are affected by a change in the pivot location. [7] have extended [5]'s study to determine the influence of pivot location on the aerodynamic response of the SD7003 airfoil for unequal pitch and plunge frequencies. From a different perspective, [8] and [9] have performed experiments on a pitching airfoil by varying the pivot location and flapping frequency. They observed that moving the pitching pivot location towards the leading or trailing edge side of the airfoil is equivalent to adding an extra component of the heaving motion to the actual pitching motion profile. [10] numerically studied the effect of pivot location on a purely pitching NACA 0012 airfoil and deduced the scaling laws of its propulsive performance (thrust, power, and efficiency) as functions of the pitching axis location and reduced frequency. Furthermore, in a recent numerical study, [11] have correlated the flow mechanism and aerodynamic response of a sinusoidal pitching airfoil for various fixed and movable pivot locations. They have proposed a new definition of the effective angle of attack to expound on the flow mechanisms observed at different pivot locations. Unlike the forced flapping airfoil studies, [12] and [13] discussed the effect of pivot locations on the wake regime and the mode of vibration of a fully passive flapping airfoil.

In all these previous studies, the thrust degradation at high flapping frequency is not resolved. This motivated us to perform the present study. In order to get enhanced thrust at $St > St_{cr}$, an asymmetricity in the pitching motion can be introduced. One way of doing that is breaking the symmetric flapping motion by incorporating two different frequencies in the downstroke and upstroke of a flapping cycle.

3. PROBLEM STATEMENT

A schematic diagram in Fig. 1 depicts the upstroke and downstroke of a symmetrically flapping elliptic airfoil. Here, the airfoil executes a sinusoidal heaving and pitching motion profile during a complete flapping cycle following Eq. (1) and (2). Also, the downstroke and upstroke frequencies (f_X) are the same in the case of symmetrical flapping. However, in our present research work, they are different. The phase angle (ϕ) difference between the heaving and pitching profile is 90° since this phase angle generates the highest propulsive efficiency [14].

$$h(t)_X = h_0 \cos(2\pi f_X t). \tag{1}$$

$$\theta(t)_X = \theta_{0X} \cos(2\pi f_X t - \phi). \tag{2}$$

In the above Eqs, h_0 = heaving amplitude = 0.75c for both downstroke and upstroke. However, θ_{0X} = pitching amplitude will vary depending upon the downstroke and upstroke frequencies. c = 0.04m is the chord length of the flapping airfoil. The incoming freestream flow speed U_∞ has been set as 0.125 m/s. f_X and ω_X are the flapping frequency and corresponding angular flapping frequency, respectively. In this study, to investigate if we can mitigate the thrust reduction at a high flapping frequency, we coupled two flapping frequencies, one from thrust reduction frequency and another one at which maximum thrust is obtained at that particular effective angle of attack amplitude (α_0). The asymmetric flapping airfoil's thrust and lift force performance is investigated at three pivot point locations, A, B and C, considered at locations 0.25c, 0.5c and 0.75c from the leading edge. The pivot points are schematically shown in Fig. 1.

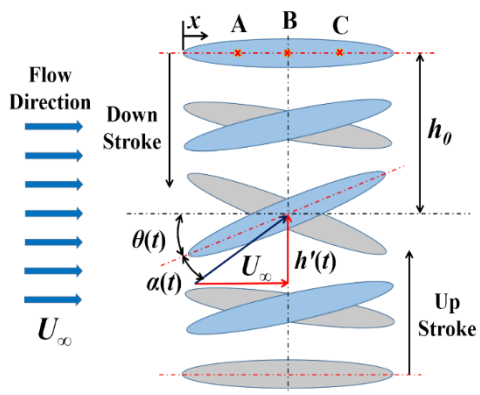

FIGURE 1: SHOWS THE FLAPPING CYCLE OF THE SYMMETRICALLY ELLIPTIC AIRFOIL. THE PIVOT POINTS A, B, AND C ON THE AIRFOIL ARE LOCATED AT 0.25c, 0.5c, AND 0.75c, RESPECTIVELY FROM THE LEADING EDGE.

We also have varied the effective angle of attack amplitudes as 10^0, 15^0, and 20^0, to assess its effects on the asymmetric flapping airfoil propulsive performance. Two non-dimensional numbers, i.e., Reynolds number, $Re = U_\infty c/\nu$, and Strouhal number $St_X = f_X A/U_{,\infty}$ are defined for our analysis. As

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f_X is different in downstroke and upstroke, corresponding St_X is also different. Here, ν is the kinematic viscosity of the fluid, and A is the width of the wake (= maximum excursion of the trailing edge of the airfoil, i.e. $2h_0$). The transient and time-averaged thrust coefficient followed in the present study is expressed in Eq. (3) and (4), respectively, where T is the thrust force, ρ is the density of the fluid, and S is the planform area of the airfoil.

T' is the physical flapping cycle time. Similarly, the transient and time-averaged C_L can be defined by replacing the thrust force with the corresponding flapping cycle lift force.

$$C_T = \left(\frac{T}{0.5\rho U_\infty^2 S} \right). \tag{3}$$

$$\overline{C_T} = \frac{1}{T'} \int_0^{T'} C_T \, dt. \tag{4}$$

4. EXPERIMENTAL SETUP

A re-circulating free-surface water tunnel with an in-house designed flapping mechanism, as shown in Fig. 2, was utilized to perform the present experiments. The elliptic airfoil considered in this investigation has a span length (L) of 0.396 m, chord (c) of 0.04 m, and a maximum thickness (t_h) of 0.005 m. The aspect ratio of the airfoil (L/c) was kept as 10, which exceeds the critical aspect ratio (≈5) value [15], so the three-dimensional effects are negligible. Moreover, to maintain the two-dimensional flow field and isolate the influence of the wingtip vortices, endplates [16] were installed at both ends of the airfoil. The incoming free stream flow speed was set at U_∞ = 0.125 m/s, such that the Reynolds number with respect to the chord of the airfoil was 5000. The flow was driven by a Thompson Kelly and Lewis 11KW centrifugal pump and a Teco speed-con 440V Class Inverter. A Butterfly valve, honeycomb, and series of 4 fine screens (square mesh) were used to smoothen the perturbations in the flow before it reached the test section. The water tunnel test section has a dimension of 0.4m × 0.4m × 1.8m. The floor and sidewalls of the test section were made of transparent perspex. The flow rate was monitored by an Endress Hauser Promag flow meter.

The elliptic wing was suspended from the four load-bearing bars of the force sensor. The MDrive23Plus micro-stepping motor drives the pitching and heaving motion of the airfoil. The resolutions of the motors were set at 0.36^0/step for translation motion with the corresponding linear translation of 0.01mm/step and pitching motion at 0.036^0/step. The required flapping motion of the elliptic airfoil was generated by two separate stepper motors operating in a synchronized operation. Additional configuration details of the stepper motors, force measurement, and post-processing of the measured data can be referred to in [16]. Note that due to limitations of the experimental facility, only experimental data are available upto flapping frequency (or St) of 0.5 for all three pivot points.

In addition to the experiments, numerical simulations were performed in the present study. The numerical solver details are discussed in the following.

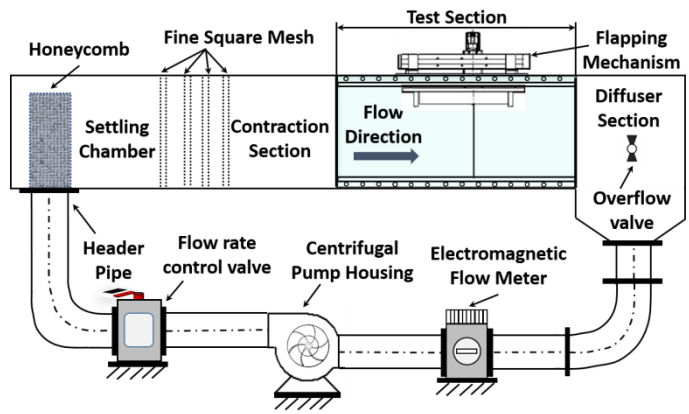

FIGURE 2: SCHEMATIC DIAGRAM OF THE RE-CIRCULATING FREE SURFACE WATER-TUNNEL SETUP.

5. NUMERICAL METHODOLOGY

In the present study, a finite volume method (FVM) based commercial software package, ANSYS FLUENT, is used to simulate the transient incompressible flow over a flapping elliptic airfoil at Re = 5000 and varied St with a laminar viscous flow model. The effectiveness of the ANSYS FLUENT solver in simulating flapping airfoil problems has been thoroughly verified and documented in the literature [16] and [17]. Using the dynamic mesh principle and following the arbitrary Lagrangian-Eulerian (ALE) formulation, the conservative form of the governing equations shown in Eq. (5) is solved in an inertial frame of reference on an arbitrary control volume cv with moving control surface cs [18].

$$\frac{\partial}{\partial t}\int_{cv} \rho\psi dV + \int_{cs} \rho \mathbf{n}\cdot\left(\nu - \nu_g\right)\psi dS - \int_{cs} \rho\Gamma_\psi \mathbf{n}\cdot\nabla\psi dS = \int_{cv} S_\psi\left(\psi\right) dV, \tag{5}$$

where, the variables ν, ν_g, Γ_ψ, S_ψ, and ρ, are the flow velocity, moving mesh velocity, diffusion coefficient, source term of ψ per unit volume, and fluid density, respectively. $\mathbf{n}$ is the unit vector normal to the control surface. Pressure and velocity coupling are done using the Pressure-Implicit with Splitting Operators [18] (PISO) algorithm. We have used the PISO skewness correction for highly-distorted meshes, and the under-relaxation factors for the momentum and pressure are defined as 0.3 and 0.7, respectively. The momentum equation's spatial and temporal discretization is carried out using the second-order upwind scheme and first-order implicit scheme [18], respectively. A double-precision solver setting is followed to approximate the computer-generated round-off errors. The

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convergence criterion for the iterative method for both momentum and mass residuals is set to O(10^{-5}).

5.1 Computational domain, mesh and solver validation

As shown in Fig. 3(a), an O-type computational domain is adopted in the present simulations. The domain is subdivided into two zones with a moving inner zone Ω_1 and a stationary outer zone Ω_2. Between the two zones, an interface boundary condition is applied. The inner (Γ_1) and outer (Γ_3) boundaries are located at a radius of 15*c* and 30*c*, respectively, so that the far-field boundary effects on the airfoil (Γ_0) are negligible. At the start of the flapping stroke, the geometric center of the elliptic airfoil is located at 0.75*c* from the origin (O) of the domain. The left half of the outer boundary (Γ_2) is inlet; whereas the right half is outlet. On the inlet section, a uniform freestream velocity (U_∞) in the X-direction is assigned. No-slip velocity condition on the airfoil surface and atmospheric pressure condition on the outlet boundary is applied, respectively. It is to be mentioned that, the two junction points where the inlet and outlet coincides, is located far away from the airfoil surface. So, it has minimal effect of the near wake flow regime. Standard initialization is adopted for computing the result from the inlet. To obtain the flapping motion profile shown in Eq. (1) and (2), a user-defined function (UDF) is written in C-programming language and compiled in ANSYS FLUENT solver.

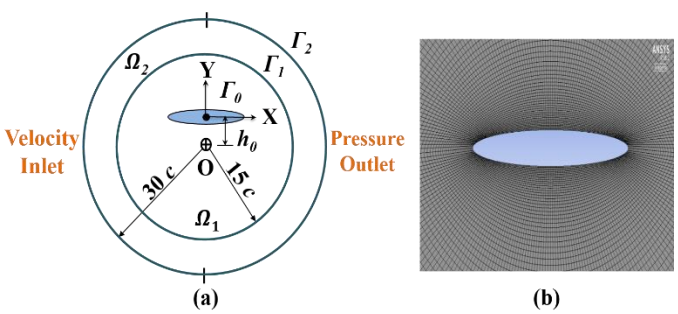

FIGURE 3: (a) REPRESENTS O-TYPE COMPUTATIONAL DOMAIN AND (b) CLOSE-UP VIEW OF THE STRUCTURED MESH NEAR THE ELLIPTIC AIRFOIL.

A fine mesh with quadrilateral structured elements is followed in this study. A close-up view of the mesh around the airfoil is shown in Fig. 3(b). The diffusion smoothing-based dynamic mesh technique was applied inside the deforming zone. As shown in Table 1, five different mesh sizes were selected to perform the mesh independence study keeping the pivot location at the airfoil geometric center (pivot B) and considering St = 0.32, Re = 5000, $\phi = 90^0$, $h_0 = 0.75c$ and $\theta_0 = 30^0$. The transient C_T plot comparison in Fig. 4(a) shows no significant variation with increasing mesh density. Moreover, the error percentage of time-averaged C_T is below 0.1% for the mesh size M4. Therefore, this mesh size has been chosen for the rest of the simulations in the present study.

We have considered a NACA 0012 airfoil following sinusoidal heaving and pitching motion to validate the present numerical solver as mentioned in Eq. (1) and (2). The pivot point is fixed at 1/3rd chord length (*c*) from the leading edge. Here, *c* = 0.01m. The O-type computational domain and the same set of boundary conditions as discussed in section 5.1 are followed. The flapping parameters are adopted from the study of [19]; Re = 4000, $\phi = 90^0$, $h_0 = 0.75c$, St is varied between 0.2 to 0.6 at an interval of 0.1. Here, $\alpha_0 = 20^0$, and corresponding θ_0 values are calculated following the steps mentioned in [15]. In Fig. 4(b), the transient C_T results obtained in the present solver are compared with that of [19]. They are found to be a good match. This further confirms the accuracy of the present solver.

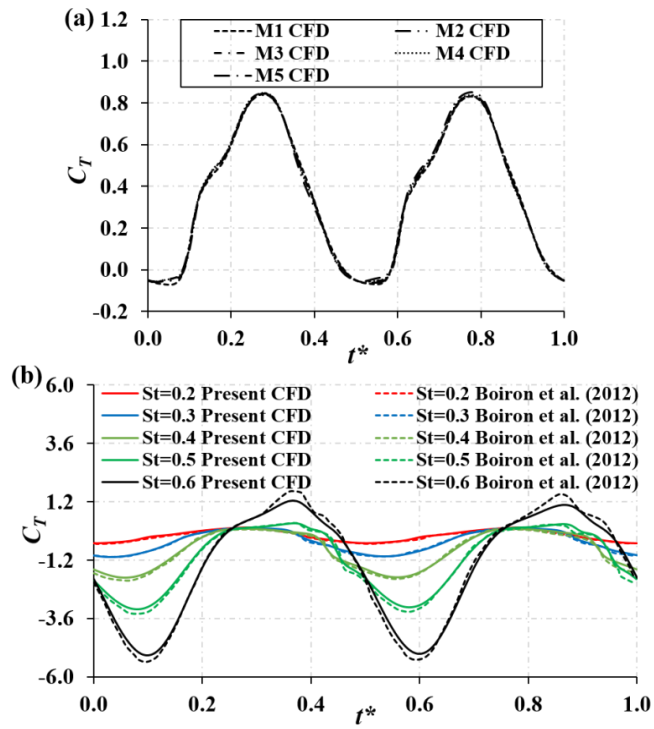

FIGURE 4: (a) SHOWS THE TRANSIENT C_T PROFILE COMPARISON FOR THE MESH INDEPENDENCE STUDY. THE TRANSIENT C_T PROFILE IN THE PRESENT SOLVER IS COMPARED IN (b) WITH THAT REPORTED IN [19].

TABLE 1: MESH INDEPENDENCY TEST

Grid name	*No of nodes*	*C_T value*	*% Error* $\left\lvert\frac{C_{Tnew}-C_{Told}}{C_{Told}}\right\rvert \times 100$
M1	29,040	0.3636	-
M2	50,960	0.3659	0.63
M3	**66,300**	**0.3663**	0.11
M4	90,090	0.3666	**0.08**
M5	1,12,230	0.3668	0.05

6. RESULTS AND DISCUSSION

In this section, we have analysed the thrust and lift performance of the asymmetrically flapping airfoil and

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associated near wake flow features at three different pivot locations ('A', 'B' and 'C'; see in Fig. 1), three effective angle of attack amplitudes (α_0) and a range of non-dimensional flapping frequencies (St = 0.4 – 0.8).

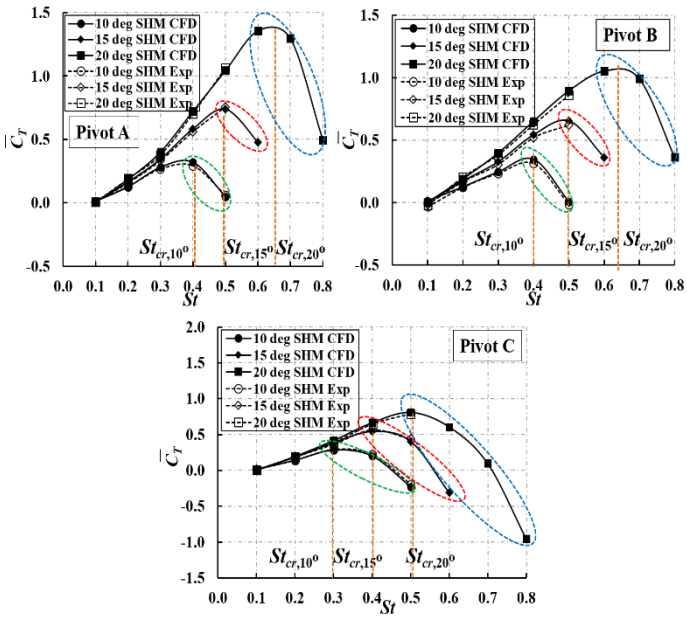

FIGURE 5: NUMERICAL AND EXPERIMENTAL TIME-AVEREGED C_T DISTRIBUTION AS FUNCTIONS OF St AND α_0 FOR PIVOT POINT A, B AND C. THRUST REDUCING REGIMES ARE MARKED IN DOTTED CIRCLES. CORRESPONDING St_{cr} ARE ALSO SHOWN BY DOTTED VERTICAL LINES.

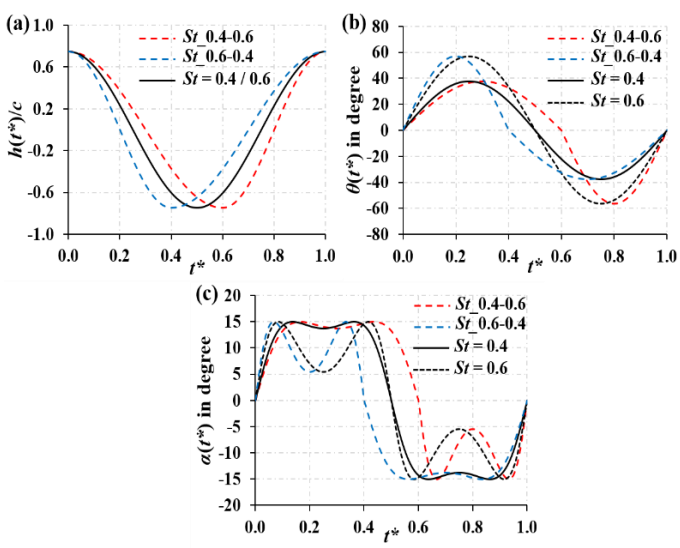

FIGURE 6: INSTANTANEOUS HEAVING, PITCHING AND EFFECTIVE ANGLE OF ATTACK PROFILES FOR THE SYMMETRIC FLAPPING AND COMBINED FREQUENCY CASES ARE SHOWN IN (a), (b) AND (c).

This is the range of St where thrust performance starts to degrade at the selected α_0 values in the case of the symmetric flapping elliptic airfoil [15, 16, 17]. In Fig. 5, the numerical and experimental time-averaged C_T distribution as functions of St and α_0 for pivot point A, B, and C are presented. The thrust reducing regimes are marked in dotted circles, which is our parametric investigation space.

In order to achieve enhanced time-averaged thrust at high flapping frequencies ($St > St_{cr}$) for a particular α_0 value, we have implemented two separate flapping frequencies during the downstroke and upstroke of the flapping cycle. More specifically, the downstroke flapping frequency is always selected with a value higher than the St_{cr}, but the upstroke flapping frequency is kept fixed as the critical Strouhal number. It is important to mention here that the sequence of St is swapped suitably to get the best aerodynamic performance. For example, if downstroke and upstroke frequencies are 0.4 and 0.6, respectively, then it's denoted as St_0.4 – 0.6. If we swap the frequencies, it will become St_0.6 – 0.4.

In Fig. 6, heaving, pitching, and effective angle of attack profiles of the present asymmetric flapping airfoil are plotted for the symmetric flapping frequencies as well as a combination of two frequencies. For a representative case, we have selected St = 0.4 and 0.6 as symmetric cases and St_0.4 – 0.6 and St_0.6 – 0.4 as combined frequency cases keeping the α_0 fixed at 15^0. The non-dimensional time scale is t^*. Here, t^* ($=t/T'$) is defined as the ratio of the computational time 't' and the physical flapping cycle time T'. Here, 't' and T' are adjusted in such a way that a smooth stroke reversal is achieved for combined frequency cases. Note that the St of the flapping airfoil is modified while keeping the peak magnitude of the effective angle of attack (α_0) fixed. In particular, we have considered three α_0, namely 10, 15, and 20 degrees; the corresponding θ_0 is evaluated by following the steps mentioned in [17] for a fixed h_0 (= 0.75c) value (refer to Fig. 6(a)). It is worth noticing that, in Fig. 6(c), at high St (for example, St = 0.6), higher-order sub-harmonics appear in the sinusoidal $\alpha(t^*)$ profiles. Additionally, from Fig. 6(b), it can be interpreted that at a high St value, to keep α_0 constant, a higher θ_0 or higher rate of airfoil rotation is needed. Previous studies such as [17] and [15] reported that this phenomenon generates an adverse suction region near the flapping airfoil, which explicitly degrades its overall thrust performance. So, combining a lower flapping frequency with the higher one during a flapping cycle can mitigate the thrust degradation by reducing or eliminating the effects of a high pitching rate or higher-order sub-harmonics. Consequently, a significant amount of thrust enhancement can be obtained along with non-zero lift performance.

6.1 Time-averaged thrust and lift performances

The following section evaluates the two-dimensional flapping elliptic airfoil's aerodynamic thrust and lifts force performance as functions of the pivot locations (namely, A, B, and C), St, and α_0 values. As shown in Fig. 5, for each pivot location, based on their effective angle of attack amplitudes, the critical St and thrust degradation regime differed. To get higher

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thrust performances at these high *St*, we combined two frequencies. One frequency is selected from the thrust reduction regime, whereas the other one is kept at critical *St*, where maximum thrust has been reported for symmetric flapping cases. The time-averaged C_T and C_L values from combined frequency cases are compared with symmetric flapping cases. It is worth highlighting that the observations are periodic and swapped suitably when the flapping stroke frequency selection is reversed in the flapping cycle.

As a result, time-averaged thrust remains the same for both frequency combinations, but time-averaged lift becomes negative in some of the swapped frequency cases.

TABLE 2: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X – Y) FLAPPING MOTION AT DIFFERENT PIVOT LOCATIONS AND α_0 VALUES. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

Pivot A, 10deg	***St***	C_T	C_L
	St = 0.5	0.0466	0.0011
	St = 0.4	0.3190	0.0024
	St_0.5 - 0.4	0.2411	0.0913
Pivot A, 15deg	***St***	C_T	C_L
	St = 0.6	0.4779	0.0005
	St = 0.5	0.7396	0.003
	St_0.6 - 0.5	0.6571	0.2080
Pivot A, 20deg	***St***	C_T	C_L
	St = 0.7	1.2959	0.0024
	St = 0.6	1.3586	0.0067
	St_0.7 - 0.6	1.3162	0.2320
Pivot A, 20deg	***St***	C_T	C_L
	St = 0.8	0.4939	0.0078
	St = 0.6	1.3512	0.0049
	St_0.8 - 0.6	1.2934	0.5730
Pivot B, 10deg	***St***	C_T	C_L
	St = 0.4	0.3455	0.0018
	St = 0.5	0.0066	0.0011
	St_0.4 - 0.5	0.1515	0.0270
Pivot B, 15deg	***St***	C_T	C_L
	St = 0.5	0.6539	0.0023
	St = 0.6	0.3604	0.0029
	St_0.5 - 0.6	0.5023	0.0220
Pivot B, 20deg	***St***	C_T	C_L
	St = 0.6	1.0531	0.0037
	St = 0.7	0.9901	0.0058
	St_0.6 - 0.7	1.0212	0.2570
Pivot B, 20deg	***St***	C_T	C_L
	St = 0.6	1.0531	0.0073
	St = 0.8	0.3633	0.0089
	St_0.6 - 0.8	0.7812	0.2020
Pivot C, 10deg	***St***	C_T	C_L
	St = 0.3	0.2831	0.0089
	St = 0.5	-0.2330	0.0047
	St_0.3 - 0.5	0.1183	0.4462
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.3	0.2833	0.0157
	St = 0.4	0.2021	0.0071
	St_0.3 - 0.4	0.2200	0.1987
Pivot C, 20deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0155
	St = 0.5	0.4044	0.0023
	St_0.4 - 0.5	0.4122	0.2066
Pivot C, 20deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0066
	St = 0.6	-0.3100	0.0037
	St_0.4 - 0.6	0.0789	0.2672
Pivot C, 10deg	***St***	C_T	C_L
	St = 0.5	0.8061	0.0079
	St = 0.6	0.5257	0.0014
	St_0.5 - 0.6	0.5989	0.3035
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.5	0.8061	0.0073
	St = 0.7	0.0929	0.0098
	St_0.5 - 0.7	0.4405	0.0103
Pivot C, 20deg	***St***	C_T	C_L
	St = 0.5	0.8062	0.0187
	St = 0.8	-0.9600	0.0122
	St_0.5 - 0.8	0.6251	0.0984

In Table 2, only those sets of values are presented where the positive time-averaged lift is observed.

We know from [17] that the time-averaged C_T value for pivot A, at any given symmetrical flapping frequency and α_0 value is always higher than pivot B and C. However, the time-averaged C_L value is found to be negligible in all symmetric flapping cases. The critical *St* for pivots A and B is similar; that's why the parametric space of investigation is also the same for them. However, pivot C is different due to the early thrust degradation. It is observed in Table 2 that for the entire parametric space which is investigated here, the time-averaged C_T is always enhanced by combining two frequencies in a flapping cycle when compared with the values obtained at $St > St_{cr}$. But at the same time, the time-averaged C_T values are slightly lesser than that of St_{cr}. Though the thrust degradation cannot be delayed to higher *St*, it at least can achieve higher thrust magnitudes at large flapping frequencies. Additionally, the non-zero time-averaged lift is obtained from this asymmetric flapping motion. Interestingly, for pivot A, a faster downstroke and slower upstroke combination is proven to be better in terms of aerodynamic performance. Whereas for pivots B and C, slower downstroke and faster upstroke are better. Maybe that's why the insects, birds and unmanned

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vehicles flap their wings, keeping the hinge point towards the leading edge. Because that's where higher thrust and non-zero lift are produced at high flapping frequencies during the faster downstroke and slower upstroke. The time-averaged and transient aerodynamic force coefficient variations are plotted for a converged and periodic flapping cycle. There is no combined *St*, which is common for all three pivot points. That's why we have taken a representative case of *St*_0.4 – 0.6, α_0 = $15^{0,}$ and pivot B to understand the aforementioned time-averaged force variations.

6.2 Time-averaged and transient aerodynamic forces for center pivot location (pivot B)

TABLE 3: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X – Y) FLAPPING MOTION AT PIVOT B AND α_0 =15^0. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

	St	C_T	C_L
Pivot B, 15deg	St = 0.4	0.5427	0.0027
	St = 0.6	0.3604	0.0005
	St _0.4 - 0.6	0.4659	0.2812
Pivot B, 15deg	***St***	C_T	C_L
	St = 0.6	0.3604	0.0127
	St = 0.4	0.5427	0.0011
	St_0.6 - 0.4	0.4622	-0.2735

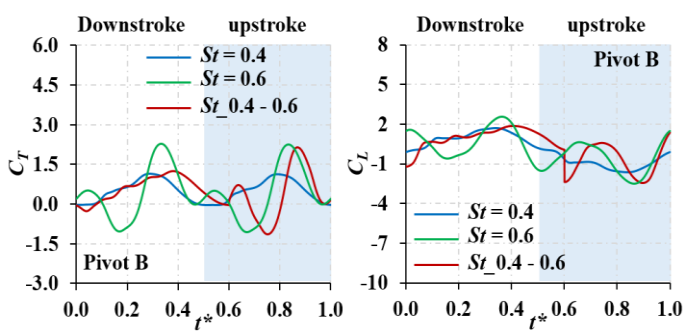

FIGURE 7: SHOWS THE COMPARISON OF TRANSIENT C_T AND C_L PROFILES BETWEEN *St* = 0.4, 0.6 AND *St*_0.4 – 0.6 CASE FOR PIVOT B.

In this paragraph, to understand the distribution of time-averaged aerodynamic forces, we plotted transient C_T and C_L profiles for the above-mentioned representative case of *St*_0.4 – 0.6 at α_0 = 15^0 and compared them with *St* = 0.4 and 0.6 cases in Fig. 7. It is observed that during the downstroke of *St*_0.4 – 0.6, the thrust reducing valley region of *St* = 0.6 got eliminated; however, the same remained in upstroke, which causes the thrust value to be higher than *St* = 0.6 but lesser than *St* = 0.4. Due to symmetric flapping motion, the time-averaged lift generated at *St* = 0.4 or 0.6 is almost zero. But for *St*_0.4 – 0.6, the transient lift profile is such that it generates a non-zero lift, as reported in Table 3. Importantly, when we swapped the frequencies in downstroke and upstroke, i.e., *St*_0.6 – 0.4, the transient thrust profile being reciprocal of *St*_0.4 – 0.6 case, the time-averaged thrust value remained the same. However, the modified transient profile at *St*_0.6 – 0.4 leads to a negative time-averaged lift (refer to Fig. 8) of the same magnitude as the *St*_0.4 – 0.6 case. So, for pivot B, slower downstroke and faster upstroke are beneficial for better aerodynamic performance.

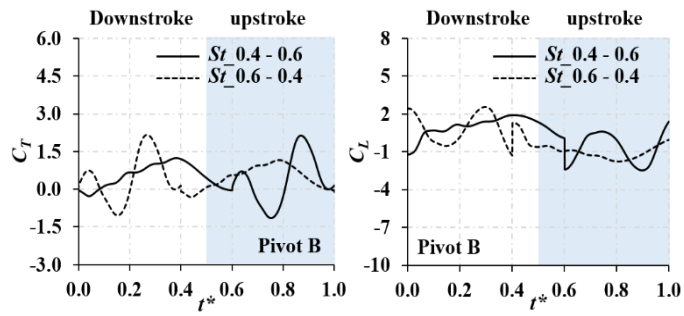

FIGURE 8: SHOWS THE COMPARISON OF TRANSIENT C_T AND C_L PROFILES FOR *St*_0.4 - 0.6 AND *St*_0.6 – 0.4 CASES OF PIVOT B.

6.3 Time-averaged and transient aerodynamic forces for off-center pivot locations (pivots A and C)

To understand the effect of pivot point locations on the time-averaged and transient aerodynamic performances, we have taken the *St*_0.6 – 0.4 case for pivots A and C.

Similar to pivot B, in the case of pivot C also, we can observe a sharp valley region in the transient C_T and C_L profiles near the onset of downstroke (see Fig. 9), which is based on *St* = 0.6. But the same valley region is eliminated from the upstroke by incorporating *St* = 0.4. As a result, the time-averaged thrust becomes higher than that of *St* = 0.6 but lesser than *St* = 0.4. Here also, slower downstroke and faster upstroke are favorable for better aerodynamic performance because, in the case of *St*_0.6 – 0.4, the deep valley in the transient C_L profile causes negative lift (refer to Table 4). On the contrary, for pivot A, faster downstroke and slower upstroke becomes useful because of upside peak formation in the transient C_L profile (refer Fig. 9 and Table 4)

Among all three pivot locations, pivot A is better in terms of thrust and lift performance at any given combined frequency. In order to understand the reason behind this superior performance of pivot A, we analysed the transient profiles through pressure and vorticity contours at a specific time instance t^* = 0.25 (see Fig. 10). At t^* = 0.25, an expanded suction region covered the entire upper surface in the case of pivot A, which is primarily not associated with any Leading Edge Vortices (LEVs) instead, it's a consequence of Asymmetric Velocity Distribution (AVD) and Previous stroke AVD (PAVD) (refer to [17]). This ADV+PAVD induced low-pressure region generates high thrust. At the same time, a high-pressure region on the bottom surface yields high lift performance (refer to Fig. 10). Consequently, large peaks are formed in the transient profiles during the downstroke (observe Fig. 9). For pivot B, smaller LEV on the upper surface and

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lesser high-pressure region on the bottom surface of the airfoil produce lesser thrust and also negative lift at *St*_0.6 – 0.4 combination. For pivot C, the shedding of LEV from the upper surface and the formation of LEV on the bottom surface, as seen in Fig. 10, produces very less thrust as well as negative lift forces.

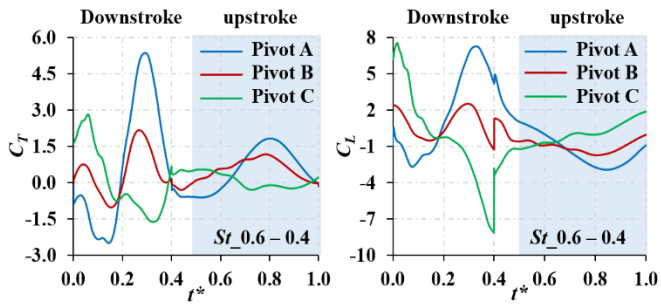

FIGURE 9: SHOWS THE TRANSIENT C_T AND C_L PROFILE COMPARISON FOR *St*_0.6 – 0.4 CASE AT PIVOT A, B, AND C.

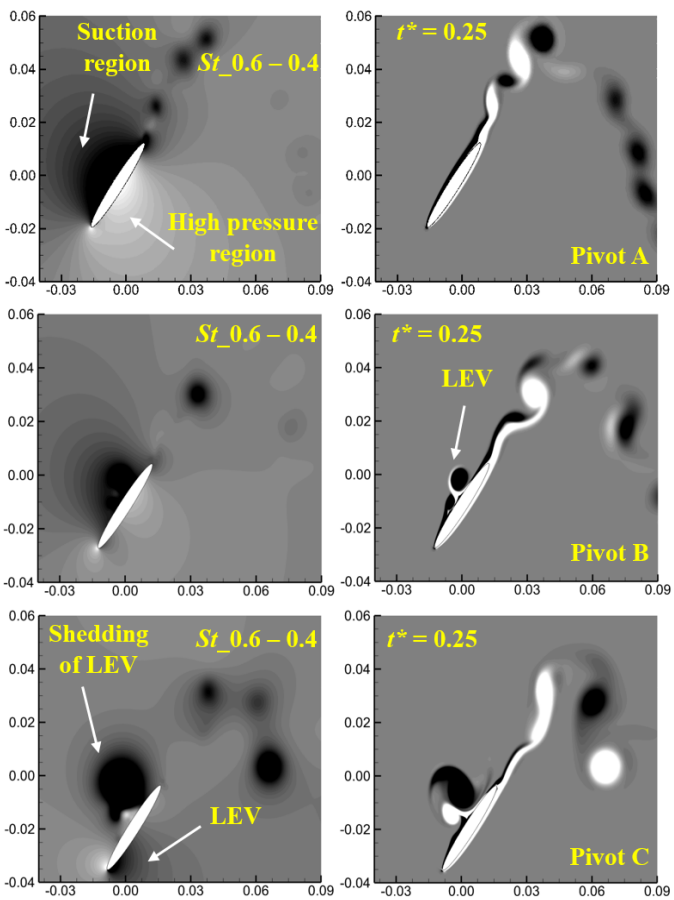

FIGURE 10: SHOWS THE PRESSURE AND VORTICITY CONTOURS FOR *St*_0.6 – 0.4 CASE AT t^* = 0.25 FOR THREE PIVOT LOCATIONS (PIVOT A, B, AND C).

TABLE 4: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X – Y) FLAPPING MOTION AT PIVOT A AND C FOR α_0 =15^0. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

	St	C_T	C_L
Pivot A, 15deg	St = 0.4	0.5827	0.0017
	St = 0.6	0.4779	0.0021
	St _0.4 - 0.6	0.6287	-0.2720
Pivot A, 15deg	***St***	C_T	C_L
	St = 0.6	0.4779	0.0037
	St = 0.4	0.5827	0.0098
	St _0.6 - 0.4	0.6247	0.2833
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0074
	St = 0.6	-0.3111	0.0099
	St _0.4 - 0.6	0.0795	0.2663
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.6	-0.3100	0.0045
	St = 0.4	0.5408	0.0142
	St _0.6 - 0.4	0.0938	-0.2528

7. CONCLUSION

In this numerical study, to mitigate the thrust deterioration at high flapping frequency (*St*) of a symmetric flapping elliptical airfoil in a forward flight condition, asymmetric flapping strokes are adopted by implementing two separate *St* during the downstroke and upstroke of a flapping cycle. Three different pivot locations at a distance of 0.25*c*, 0.50*c*, and 0.75*c* from the leading edge of the airfoil and three α_0 values of 10^0, 15^0, and 20^0 are considered for our investigation. The *St* is varied between 0.4 to 0.8. In general, the downstroke flapping frequency is higher than the critical *St*, but the upstroke flapping frequency is kept fixed as the critical *St*. The transient and time-averged results are periodic and swapped suitably when the flapping stroke frequency selection is reversed in the flapping cycle. Asymmetric flapping stroke configurations yield more time-averaged thrust and lift than symmetric flapping stroke. Interestingly, pivot A produces positive lift with faster downstroke and slower upstroke. It's the opposite for the pivot B and C. To understand these enhanced aerodynamic performances at high *St*, we have analyzed the transient thrust and lift force profiles and corresponding flow structures for all the pivot locations through pressure and vorticity contours. Pivot A is proven to be better in all aspects for symmetric as well as asymmetric flapping cases.

ACKNOWLEDGEMENTS

The authors are immensely grateful to **IIT Kharagpur** for computational support, **SERB** (Science and Engineering Research Board), and **ARDB** (Aeronautics Research and Development Board) for financial support.

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